Light and Photosynthesis

Teacher guide | Biology 2e, Chapter 8 | 45-60 minutes

Learning targets and timing

How does light level affect the time it takes leaf disks to float?

Predict how infiltration and light conditions might affect the leaf disks. Then prepare your sample, adjust the lamp, and record your observations. For a controlled comparison, keep each trial at its assigned light level for 20 minutes; you can also change the lamp during a trial and examine the exploratory results.

Suggested sequence: 8-10 min prediction and setup; 20-30 min observation; 12-20 min analysis. Virtual time can be accelerated. For photosynthesis and yeast, pool independent replicates from class groups; three sequential repeats need extra class time.

Route choices and sharing

The student pages open as guided virtual investigations. Teacher customization is deferred to a future teacher workspace. Use this worksheet to assign classroom components separately; discuss data-source differences when comparing results. Existing hybrid links still work, but the student page no longer creates them.

Materials

- Fresh spinach leaves; hole punch; three clear cups; needle-free syringe

- 0.2% sodium bicarbonate solution (0.2 g per 100 mL water), trace of dish detergent

- LED light, foil or opaque cover, timer, thermometer; 10 disks per trial

Classroom procedure

1. Make equal-sized leaf disks, avoiding major veins. Prepare matched cups of solution and label dark, low light, and bright light.

2. In a needle-free syringe, immerse disks in solution, expel air, cover the tip, and gently draw back the plunger. Gently release the vacuum and repeat as directed by your teacher. Observe and record any changes. Use the same preparation for every treatment.

3. Put 10 prepared disks in each cup. Use a cover for darkness and a shade screen for lower light; keep depth, leaf source, and temperature matched. The virtual percentages are relative settings, not calibrated classroom illuminance.

4. Count floating disks at 0, 2, 4, ... 20 minutes. Briefly observe the dark treatment with minimal illumination. Record counts even if they decrease or fail to reach five.

5. Repeat with new disks for trials 2 and 3. Independent cups, rather than individual disks in a shared cup, are the experimental replicates. Pool independent trials from class groups for a single-period lab; sequential repeats require additional time.

Safety: Use a needle-free syringe and a cool LED light. Keep electrical equipment dry. Do not taste solutions. Teacher supervision is needed for hole punches and glassware.

Light and Photosynthesis

Teacher guide / interpretation | Biology 2e, Chapter 8 | 45-60 minutes

Expected reasoning and troubleshooting

Virtual expectation: bright light generally reaches ET50 sooner than low light; darkness normally does not reach it during this model run. Exact times vary. Do not grade classroom results against the simulation.

Students should connect light-driven oxygen production and buoyancy, and recognize that respiration and leaf preparation influence net flotation.

If disks float at time zero, recheck infiltration. If none rise, check leaf condition, light, temperature, bicarbonate preparation, and excess detergent. Keep unexpected real observations; do not replace them with expected values.

Model boundaries

The interactive assay tracks an illustrative gas balance: light-driven production, continuing respiration, and gas loss. Rates, three preparation cycles, and disk variation are teaching assumptions, not calibrated predictions for a real leaf or lamp. Temperature and bicarbonate remain fixed. Guided trials require preparation and the assigned fixed light level. After all nine trials, separate exploration permits changing light intensity without replacing comparison data.

ET50 is the interpolated time when 5 of 10 disks float. Lower ET50 suggests faster net oxygen accumulation under matched conditions. If five disks never float, report "not reached within 20 minutes"; do not treat that as a zero rate.

Suggested assessment - 12 points

Dimension	Evidence to look for	Points
Design	Reasoned prediction; variables, controls or sampling plan	0-3
Data	Complete observations, correct units, source labels, repeats	0-3
Analysis	Appropriate calculations and comparisons supported by data	0-3
Explanation	Biological reasoning, uncertainty, and a feasible improvement	0-3

For each dimension: 0 = absent; 1 = partial or major errors; 2 = mostly sound with a minor omission; 3 = complete and well supported. Suggested teacher rubric only; the app does not assign a grade.

Downloads, access, and classroom pilot

Student worksheet: preparation, investigation, and exploration pages where included, plus companion cutouts. Teacher guide: procedures, interpretation, and rubric. Reports and CSV data download from Analyze & explain. Teacher guides are public resources, not protected answer keys. Before classroom use, pilot the equipment, timing, and reading level. Editable Word and fillable PDF versions are planned.

Reading and method connections

University of Arizona, floating leaf disk assay: Open source

OpenStax Biology 2e, 8.2: Open source

College Board, Biology Investigative Labs, Investigation 5: Open source

The investigation materials are original; appended cutouts were supplied by the classroom teacher. Simulated observations are not measurements from the linked sources.

Light and Photosynthesis

Teacher guide / interactive lamp investigation | Biology 2e, Chapter 8 | 45-60 minutes

Teach the preparation before interpreting flotation

Infiltration removes trapped air and replaces it with solution. Without this baseline, floating disks do not establish oxygen production. Demonstrate a needle-free syringe: expel excess air, cover the tip, gently pull the plunger, then uncover and release the vacuum. Repeat only as needed; avoid damaging leaf tissue. The simulation uses three preparation cycles as a teaching simplification.

Use controlled trials and exploratory trials

The guided comparison uses three trials at each assigned light level. Prepare the disks, add them to the cup, and check the highlighted lamp setting. The assigned setting stays fixed during each 20-minute comparison trial. Pause, speed and time-advance controls remain available. Click Collect data and continue to save the trial and advance.

After all nine trials, Explore other settings opens a separate run. Students can change lamp intensity while time advances and observe delayed changes in flotation, up to 120 simulated minutes. Exploration does not replace comparison trials. Export observations before resetting a run. No lamp-history list is displayed in the student workspace.

Suggested extension

Ask students to predict the effects of switching from high light to darkness and back. Gas remaining in the disks creates a delayed flotation response. Respiration and gas loss continue in darkness. Have students distinguish an immediate change in light from the later change in disk buoyancy.

Assessment and limits

Assess explanations of infiltration, timing, source labels, and controlled versus changed settings. Do not require real leaves to match model times. Temperature, bicarbonate, leaf health, and physical disturbance are not adjustable in this version. Switching trials, navigating away, or reloading pauses the clock; there is no background time simulation.

Cut & sort: matter and energy

Optional introduction or extension | 10-15 minutes | Biology 2e, chapters 7 and 8.

Choose Card Activity on the lab overview to open the separate circular photosynthesis and respiration model, or arrange the printed cards on chart paper. Use two molecule sets for both halves at once, or reuse one 19-card set as you switch processes.

Print cutout pages in color, single-sided, at 100% scale. Cut along dashed lines. Use scissors safely. Group work: one person arranges, one counts atoms, and one explains; then rotate roles.

Model photosynthesis. Sort all the molecule cards into reactants and products. Add plus signs and the arrow, and decide where light and chlorophyll belong.

Count carbon, hydrogen, and oxygen atoms separately on each side. Explain how your arrangement shows conservation of matter and the transfer of energy. Energy symbols are annotations, not part of this simplified atom inventory.

This is an overall net equation, not a sequence of reaction stages. The glucose diagram shows an atom inventory, not a molecular structure.

Teacher use: assign this before prediction or after analysis. Accept either order of molecule groups on the same side. Ask students to explain placement, not just match a key. The online model checks each group and atom counts for practice feedback only; no grade is submitted.

The supplied originals follow, including answer keys. Student packets omit the photosynthesis teacher page and the answer text under the respiration cutouts. For respiration, the ATP symbol does not specify yield; ADP and phosphate are not shown.

PHOTOSYNTHESIS • CUT & SORT

Cut on dashed lines. Keep all three sheets together.

Build a balanced equation. Place light and chlorophyll where they help.

O_2



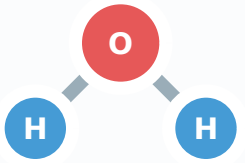
Oxygen

CO_2



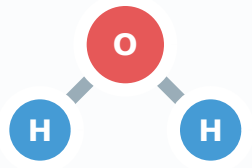
Carbon dioxide

H_2O



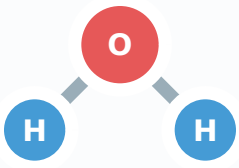
Water

H_2O



Water

H_2O



Water

O_2



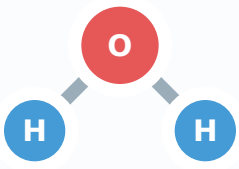
Oxygen

CO_2



Carbon dioxide

H_2O



Water

CO_2



Carbon dioxide

Atom colors: C = gray • O = red • H = blue

Student cutouts 1/3 • Print in color, single-sided, at 100%.

PHOTOSYNTHESIS • CUT & SORT

Cut on dashed lines. Keep all three sheets together.

Build a balanced equation. Place light and chlorophyll where they help.

+

Combine with

O_2



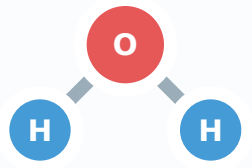
Oxygen

CO_2



Carbon dioxide

H_2O

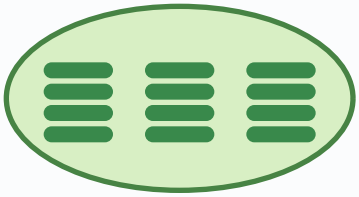


Water

+

Combine with

CHLOROPHYLL



Green pigment in chloroplasts
Simplified chloroplast diagram

Light-absorbing pigment

$C_6H_{12}O_6$



Atom inventory (not structure)

Glucose

O_2



Oxygen

LIGHT ENERGY



Sunlight

Atom colors: C = gray • O = red • H = blue

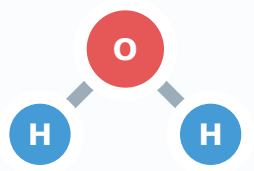
Student cutouts 2/3 • Print in color, single-sided, at 100%.

PHOTOSYNTHESIS • CUT & SORT

Cut on dashed lines. Keep all three sheets together.

Build a balanced equation. Place light and chlorophyll where they help.

H_2O



Water

→

Reaction arrow

O_2



Oxygen

O_2



Oxygen

CO_2



Carbon dioxide

CO_2



Carbon dioxide

Atom colors: C = gray • O = red • H = blue

Student cutouts 3/3 • Print in color, single-sided, at 100%.

TEACHER GUIDE

Photosynthesis: matter in, matter out, energy transferred

Copy: Print pages 1–3 for each group. Page 4 is for the teacher. Each complete set has 6 CO_2, 6 H_2O, 6 O_2, 1 glucose, 1 sun/light-energy card, 1 chlorophyll diagram, 2 plus signs, and 1 arrow.

Materials: Scissors, large chart paper, markers, and optional glue. Use sunlight or a lamp as a demonstration light source.

Student task: Sort the molecules into inputs and outputs. Use every molecule card to arrange a balanced photosynthesis equation. Place plus signs between molecule groups and the arrow between reactants and products. Place the sun/light-energy card and chlorophyll diagram above the arrow. Label the two sides “Reactants” and “Products.”

Answer key:
$6\ CO_2 + 6\ H_2O \rightarrow C_6H_{12}O_6 + 6\ O_2$
Light energy is absorbed by chlorophyll; show these above the arrow.

Check conservation: On each side, count 6 carbon atoms, 12 hydrogen atoms, and 18 oxygen atoms. The glucose picture is an atom inventory, not a structural model. Molecule pictures are simplified.

Short teacher script:
“Chlorophyll is a green pigment in chloroplasts that absorbs light energy.”
“Photosynthesis uses carbon dioxide and water to make glucose and oxygen.”
“Light provides energy. Chlorophyll helps absorb it; chlorophyll is not used up as a reactant.”
“Atoms are rearranged. Every atom on the reactant side must be accounted for on the product side.”
“Some of the absorbed light energy becomes chemical energy stored in glucose.”

Light-energy note: The single sun card represents the light energy needed for photosynthesis. Light energy is transferred and transformed; it is not an atom or a molecule. Do not include the sun or chlorophyll in the atom count.

Quick discussion: Which cards represent matter? Which represent an energy source or an energy transfer? Why is chlorophyll placed above the arrow? How does the model show that matter is conserved?